

Big Data

Apuntes

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1. Frameworks and tools

Specialized tools and frameworks are required for big data analysis when:

1. The volume of data involved is so large that is difficult to store, process and analyze data on a single machine.
2. The velocity of data is very high and the data needs to be analyzed in real-time.
3. There is a variety of data involved, which can be structured, unstructured or semi-structured, and is collected from multiple data sources.
4. Various types of analytics need to be performed to extract value from the data such as descriptive, diagnostic, predictive and prescriptive analytics.

Big Data tools and frameworks have distributed and parallel processing architectures and can leverage the storage and computational resources of a large cluster of machines.

Analytics involve several steps starting from data cleansing, data munging (or wrangling), data processing and visualization.

Specialized tools and framework are required to ingest the data from different sources into the big data analytics backend.

Based on the analysis requirements (batch or real-time), and type of analysis to be performed (descriptive, diagnostic, predictive, or prescriptive) specialized frameworks are used.

Big data analytics is enabled by several technologies such as:

- cloud computing.
- distributed and parallel processing frameworks.
- non-relational databases.
- in-memory computing.
- etc.

2. Characteristics of Big Data

2.1. Volume

Big data is form of data whose volume is so large that it would not fit on a single machine therefore specialized tools and frameworks are required to store process and analyze such data.

2.2. Velocity

- Velocity of data refers to how fast the data is generated and the primary **reason for the exponential growth of data**.
- Results in the volume of data accumulated to become very large, in short span of time.
- Some application can have strict deadlines for data analysis and the data needs to be analyzed in real-time.
- Specialized tools are required to ingest such high velocity data into the big data infrastructure and analyze the data.

2.3. Variety

Flexible enough to handle such Variety on the forms of the data:

- structured
- unstructured
- semi-structured
- text data
- image
- audio
- video
- sensor data

2.4. Veracity

Veracity refers to how accurate is the data.

- To extract value from the data, the data needs to be cleaned to remove noise.
- Cleansing of data is important so that incorrect and faulty data can be filtered out.

2.5. Value

Value of data refers to the usefulness of data for the intended purpose. The end goal of any big data analytics system is to extract value from the data.

2.6. Data collection

Before the data can be analyzed, the data must be collected and ingested into a big data stack.

The choice of tools and frameworks for data collection depends on the source of data and the type of data being ingested.

2.7. Data preparation

Data can often be dirty and can have various issues that must be resolved before the data can be preprocessed, such as:

- corrupt records.
- missing values.
- duplicates.
- inconsistent abbreviations.
- inconsistent units.
- typos.
- incorrect spellings.
- incorrect formatting.

involves various tasks such as:

- data cleaning.
- data wrangling or munging.
- de-duplication.
- normalization.
- sampling.
- filtering.

Data cleaning detects and resolves issues such as:

- corrupt records.
- records with missing values.
- records with bad formatting.
- for instance.

Data wrangling or munging deals with transforming the data from one raw format to another (Some file may be using comma as the field separator, other may be using tab as the field separator).

Normalization is required when data from different sources uses different units or scales or have different abbreviations for the same thing.

Filtering and sampling may be useful when we want to process only the data that meets certain rules.

2.8. Analysis Types

Analysis types and the algorithms.

2.9. Analysis Modes

To determine the analysis mode, which can be either batch, real-time, or interactive.

The choice of the mode depends on the requirements of the application.

If your application demands results to be updated after short intervals of time (say every few seconds), then real-time analysis mode is chosen.

If your application only requires the results to be generated and updated on larger timescales (say daily or monthly), then batch mode can be used.

If your application demands flexibility to query data on demand, the interactive mode is useful.

Once you make a choice of the analysis type and the analysis mode, you can **determine the processing pattern** that can be used.

The choice of the analysis type, analysis mode, and the data processing pattern can help you in shortlisting the right tools and frameworks for data analysis.

2.10. Visualizations

The choice of the visualization tools, serving databases and web frameworks is driven by the requirements of the application.

Static visualization are used when you have the analysis results stored in a serving database and you simply want to display the results.

If your application demands the results to **update regularly**, then you would require **dynamic** visualizations.

If you want your application to accept inputs from the user and display the results, then you would require **interactive** visualizations.

2.11. Raw Data Sources

Before the data is processed and analyzed, it must be captured from the **raw data sources** into the big data systems and frameworks.

Raw big data sources include:

- **Logs** generated by web applications and servers which can be used for performance monitoring.
- **Transactional Data:** Transactional data generated by applications such as eCommercerce, Banking and Financial.
- **Social Media:** Data generated by social media platforms.
- **Sensor Data:** Sensor data generated by Internet of Things(IoT) systems.
- **Clickstream Data:** Clickstream data generated by web applications which can be used to analyze browsing patterns of the users.

- **Surveillance Data:** Sensor, image and video data generated by surveillance systems.
- **Healthcare Data:** Healthcare data generated by Electronic Health Record (EHR).
- **Network Data:** Network data generated by network devices such as routers and firewalls.

3. Data Access Connectors

Include tools and frameworks for collection and ingesting data from various sources into the big data storage and analytics frameworks.

The choice of the data connectors is driven by the type of the data source.

Publish-Subscribe Messaging:

- Publish-Subscribe is a communication model that involves publishers, brokers and consumers.
- Publishers are the source of data.
- Publishers send the data to the topics which are managed by the broker.

Source-Sink Connectors: Source-Sink connectors allow efficiently collecting, aggregating and moving data from various sources into a centralized data store.

Database Connectors: Database connectors can be used for importing data from relational database management systems **into big data storage** and analytics frameworks for analysis.

4. Big Data

Una DB es una colección de datos organizados para facilitar y acelerar su acceso, un formato estandarizado.

La interfaz entre la DB, sistema de gestión de base de datos (DBMS).

Los DBMS están clasificados de acuerdo a varios aspectos.

- el **modelo de datos** se aplica.
- el **lenguaje de consultas** que se provee.
- el **hardware** que se requiere.
- el **diseño interno**
- su **desempeño y escalabilidad**.

El modelo de datos define la estructura lógica, el modelo de datos estandariza cómo los elementos de datos se relacionan entre sí.

Las DB pueden ser consideradas como: navegacionales, racionales y no relacionales (NoSQL)

4.1. Navegacionales Desuso

Cuando las DB se almacenaban en cintas magnéticas, se necesita navegar a través de los datos.

Las referencias a la ubicación incluida en estructuras de datos básicas.

4.2. DB Relacionales

Es una descripción matemática de teoría de conjuntos llamada álgebra relacional.

Se usa el lenguaje estructurado de consultas **SQL** para interactuar con este tipo de DB.

Los datos están organizados a través de relaciones compuestas de tuplas las cuales mapean atributos a valores atómicos de longitud limitada.

Las transacciones si tienen características como ACID:

- **Atomicidad** si la transacción falla de la DB es devuelta a su estado anterior (rollback).
- **Consistencia** una DB es llevada de un estado consistente a otro estado consistente. Los estados intermedios pueden ser no consistentes.
- **Aislamiento** las transacciones concurrentes no se afectan entre si.
- **Durabilidad** los cambios realizados por una transacción se vuelven permanentes.

4.3. NoSQL

El uso masivo de las aplicaciones web dio como resultado nuevos requerimientos para las bases de datos.

No pueden ser satisfechos por las DB convencionales.

El término NoSQL se usa de forma genérica para todas las DB que se diferencian de las DB relacionales.

Características:

- No se requiere de un esquema fijo de tablas y registros, se pueden tener varios modelos: llave-valor, basado en columnas, basado en grafos y basado en documentos.
- Escalamiento horizontal, la mayoría de estas DB permiten el particionamiento de los datos y su procesamiento sobre varios nodos.
- Soporte de multiprocesadores: diseñados para trabajar en máquinas multiprocesador, al agregar mas procesadores se puede obtener un **escalamiento lineal en el desempeño**.
- Eliminación de operaciones de unión (join).
- Soporte modelos débiles de concurrencia, muchas de las DB NoSQL **relajan aspectos de las transacciones ACID**.

- Son DB sin esquemas, **los datos se almacenan sin un esquema fijo**, se pueden agregar nuevos campos a los registros.
- Arquitectura **compartir-nada**, muchas NoSQL son diseñadas para ser funcionales en clúster, donde no se comparte memoria ni discos.

4.4. NoSQL Características

El surgimiento del BD es muy demandante para las DB.

Algunos requerimientos exceden las capacidades de las DB tradicionales.

Accesibilidad, disponibilidad, y balanceo de cargas en servidores distribuidos geográficamente.

Propiedades en un sistema distribuido de datos:

- **Consistencia:** propagación de la misma actualización de los datos a todos los nodos. Se garantiza la consistencia de datos en todas las particiones replicadas de a DB.
- **Disponibilidad:** Debe estar disponible, aún en casos de falla de comunicación, gracias a las replicas de la DB.
- **Tolerancia** de las particiones, habilidad de dos segmentos del sistema (particiones) de **proceder ante la presencia de fallas** de comunicación. Mientras más tolerancia de fallas se tenga, la DB esta mas disponible, sin embargo, la accesibilidad de datos es menos consistente y viceversa.

Es **imposible** asegurar tres propiedades al mismo tiempo en un sistema distribuido de datos (máximo dos de ellas).

El **balance** en el ajuste de esas propiedades depende de los requerimientos de la aplicación.

Toda aplicación de BD en un sistema distribuido de datos tiene que hacer un acuerdo entre la disponibilidad y consistencia.

Para la mayoría de las aplicaciones web la disponibilidad es crucial.

No disponibilidad o aplicaciones lentas puede provocar pérdida de clientes. Las aplicaciones web **no son transaccionales**.